CHAPTER

ATA 51

STANDARD PRACTICES & STRUCTURES

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Contents

Introduction

Most of the Dash 8 Series 400 aircraft structure is made from aluminium alloys. Alloy steels, corrosion-resistant steels, titanium, organic- fibre laminates and composites are used where necessary for strength, long serviceable life or heat protection. To decrease weight, honeycomb structures and chemically-milled panels are used in many locations.

General

The airframe structure is made to be damage-tolerant against damage that results from fatigue, corrosion or accidental damage during the life of the aircraft. After damage has occurred (which the flight or maintenance crews cannot easily see), the remaining serviceable structure will be sufficiently strong to transmit usual operational loads until the damage is found and repaired without:

- Dangerous failure
- Too much structural deformation.

This is done by the usage of damage-resistant materials and multiple load-path construction together with scheduled structural inspection. The inspection frequency is found by damage tolerance analysis.

Overview

The aircraft is a metal high winged monoplane with fully cantilever wings and horizontal stabilizer surfaces, a semi-monocoque fuselage and a fully retractable tricycle landing gear. Power is provided by two turbo-prop engines in wing mounted nacelles. A large portion of the skin panels are bonded assemblies consisting of a skin, stringers and doublers, or skin sandwich with a honeycomb core.

Refer Figure 51-1

The aircraft structure is divided into primary and secondary classes to show structural importance. The classification of the structure will help you make decisions about how and when to repair damage. The primary and secondary structural identity of fittings and attachments is the same as the structures to which they are attached.

Primary Structure: The function of primary structure is to transmit loads that are caused by:

- Pressurization
- Wind gusts
- Flight manoeuvres
- Take-off and landings
- Ground manoeuvres.

The primary structure is divided into structural significant items (SSIs). The structural significant items are those structural parts that are the most important in the transmission of flight, ground and pressurization loads. Failure in the structural significant items of the primary structure can possibly cause structural collapse, loss of control or loss of motive power.

NOTE: When repairs are made to structural significant items, you must make sure that the repair is damage- tolerant.

It is possible that all or part of a SSI is classified as an Airworthiness Limitation Item (ALI). The ALIs are those parts of the structure where mandatory maintenance and inspection are necessary to prevent the loss of airworthiness due to fatigue cracking. Some ALIs are replaced before fatigue cracks can occur. The remaining ALIs are inspected for cracks with a time controlled inspection program. The ALIs are identified in the Maintenance Program Document.

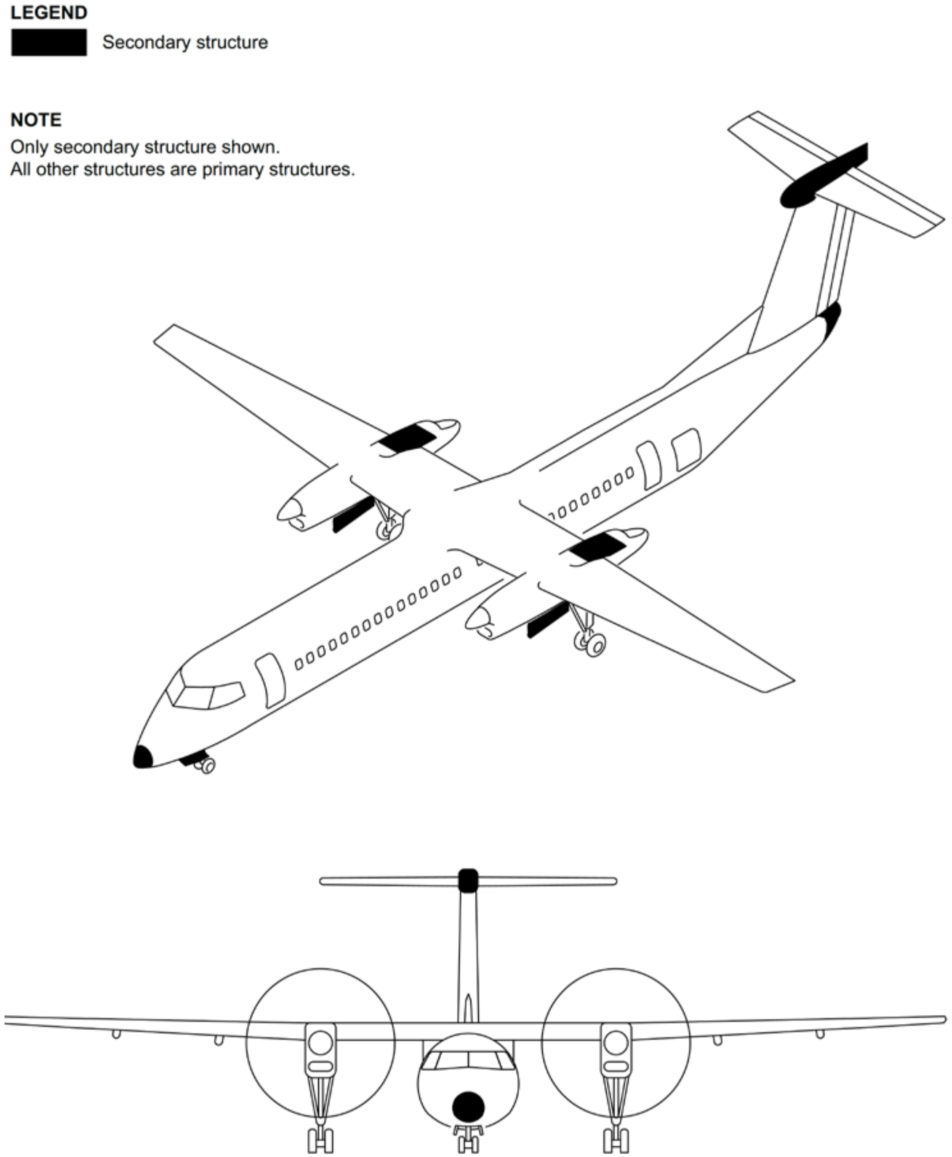


Figure 51-1 Aircraft Secondary Structure

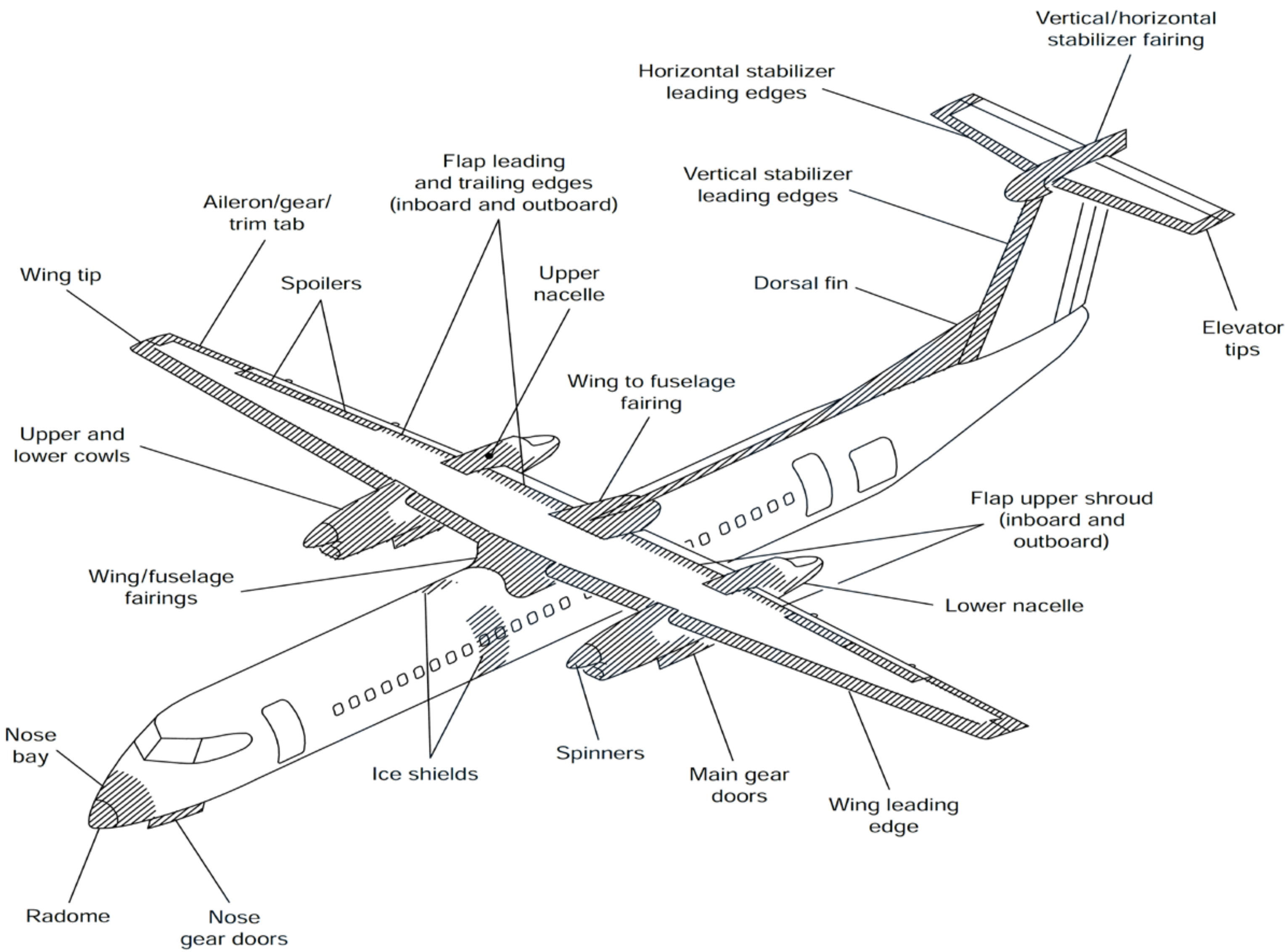


Figure 51-2 Aircraft Composite Structures

Secondary Structure: The primary structure can contain auxiliary parts that are secondary structure. Secondary structures are the ones that

- Help the primary structure transmit aerodynamic loads
- Keep the aerodynamic shape
- Attach equipment.

Loads applied to the secondary structure are transmitted to the primary structure. Damage to the secondary structure can possibly cause unwanted effects to the primary structure, engine performance, aerodynamic qualities and/or surface stiffness. A careful analysis must be made of damage to secondary structure for these unwanted effects.

Refer Figure 51-2
A large portion of skin panels, leading edges, shrouds, fairings, ice shields and other structures are made of composite materials.

Do not use standard practices for any composite repairs to the aircraft. Refer to the applicable Chapter, Section, Subject in Structural Repair Manual, PSM 1-84-3 for specific repairs to composites or contact the Manufacturer.

Training Information Point

Damage is a change in cross-sectional area, separation (fully or partially), or permanent distortion of a structural part.

Damage to the aircraft can be caused by collision, fatigue, fire or corrosion.

Damage can also be caused by operation of the aircraft at a higher stress level permitted by engineering limits.

Damage is divided into three distinct classifications:

- Allowable damage
- Repairable damage
- Damage repaired by replacement of affected parts.

Allowable damage: This is defined as light damage in the form of dents, nicks, scratches or surface corrosion, the extent of which has no measurable effect on the required strength and durability of the component, its aerodynamic qualities and intended function or performance. Such damage must be free from cracks.

Corrective procedures or repairs are not necessary. Only small corrective procedures or repairs are necessary, such as:

- Removal of nicks and scratches
- Removal of light surface corrosion
- Corrosion preventive treatment.

Repairable damage: This is defined as damage exceeding the limits of allowable damage and maybe repaired by one of the types of repair defined in each chapter, generic repair manual or by a specific approved repair scheme supplied by the manufacturer.

Repair by part replacement: Replacement of parts is in general the most satisfactory repair procedure. When the repair is correctly done, the area has the same strength, serviceable life, weight and configuration as it had before it was damaged. This can also be more economical than the repair of large areas of damage. Replacement of parts is necessary when:

- Parts are damaged and not economical to repair
- Damaged parts are too small for alternate means of repair
- Forged, cast and machined fittings are damaged more than the allowable damage limits
- After a repair to highly-stressed parts, the allowance for safety will not be sufficient
- Parts that have their strength decreased by the effect of fire or heat, metal becomes soft or weakened.

The following are definitions to the type of damage, given as an aid to damage identification and classification.

Dent: A dent is normally a damage area which is depressed or deformed with respect to its normal contour. The area boundaries are smooth and uncracked. There is no change in the cross-sectional area of the material.

Abrasion: An abrasion is a damage area of any size which results in a sectional area change in the material due to scuffing, rubbing, scrapping or other surface erosion.

Gouge: A gouge is a continuous sharp or smooth groove in the surface of the material. A loss of material or cross-sectional area is associated with a gouge.

Nick: A nick is a localized gouge with sharp edges. A series of nicks in a line pattern are considered to be equivalent to a gouge

Scratch: A scratch is a line of damage, usually caused by contact with a very sharp object and does not penetrate the full thickness of the material. A scratch which penetrates the full thickness is considered a crack.

Crack: A crack is a partial or complete fracture in the material and is usually in the form of an irregular line.

Puncture: A puncture is a localized hole or perforation which penetrates the full thickness of the material.

Corrosion: Corrosion is a damage area due to exposure of unprotected metal surfaces environmental conditions or due to electrochemical action. Depth of such damage must be determined after removal of the corrosion.

Void: Local absence of bonding between any two bonded elements or local absence of core in a sandwich structure.

Wrinkle: Local waviness of a surface.
Buckle: A surface that is warped or distorted.

Crease: A depression in the surface of material caused by contact with another object.

Delamination: Separation of layers in a bonded assembly.

Safety Precautions

- Use approved protection equipment such as rubber gloves, safety glasses, rubber apron and respirators.
- Avoid skin contact with solvents or sealants.
- Should accidental eye contact with solvents/ adhesives occur, flush immediately with large quantities of water and seek immediate medical attention.
- Keep your work area clean and have a good flow of clean air.
- Provide sufficient ventilation when working in confined spaces, avoid prolonged breathing of solvent and sealant fumes.
- Keep CPC, solvents and sealants away from fire and other sources of ignition.
- No sparks, open flames or heat lamps in the work area.
- Ground the aircraft, all the major components, and the equipment.
- Do not eat or drink, or smoke in areas where solvents are being handled.
- If ingestion of solvent occurs, get medical aid immediately.
- If irritation occurs, get medical aid. Solvents are poisonous.
- Refer to the applicable manufacturer’s Material Safety Data Sheet (MSDS) for specific data on any of the materials used.

Environment Protection

Many of the materials used can cause damage to the environment. When you dispose of these materials, you must follow the local government health and safety regulations.

Critical Zones

Refer Figure 51-4

This section specifies the critical zones in the Dash 8, Series 400 aircraft. Any deviation or repairs must be approved by the manufacturer.

Aircraft Region	Critical Zones
Non-Moveable Surfaces	
Wing	All non−moveable wing upper and lower surfaces are in the critical zone.
Nacelle	All areas ahead of the firewall (including propeller, spinner and intake duct) and areas within 1 foot of the surface of the wing.
Fuselage	The fuselage surface (including fairings), ahead of the wing rear spar. Note also that the airstair door has a less demanding tolerance.
Vertical and horizontal stabilizer	The entire surface is in the critical zone.
Vertical and horizontal stabilizer fairings	The entire surface is in the critical zone.
Dorsal fin	The entire surface is in the critical zone.
Moveable Surfaces	
Aileron (including tabs)	The entire surface is in the critical zone.
Flaps	Figure 51-3, illustrates the critical and non−critical regions for a single element.
Spoilers	The entire surface is in the critical zone.
Rudder and elevators (including tabs)	The entire surface is in the critical zone.

Table 51-1 Critical Zones

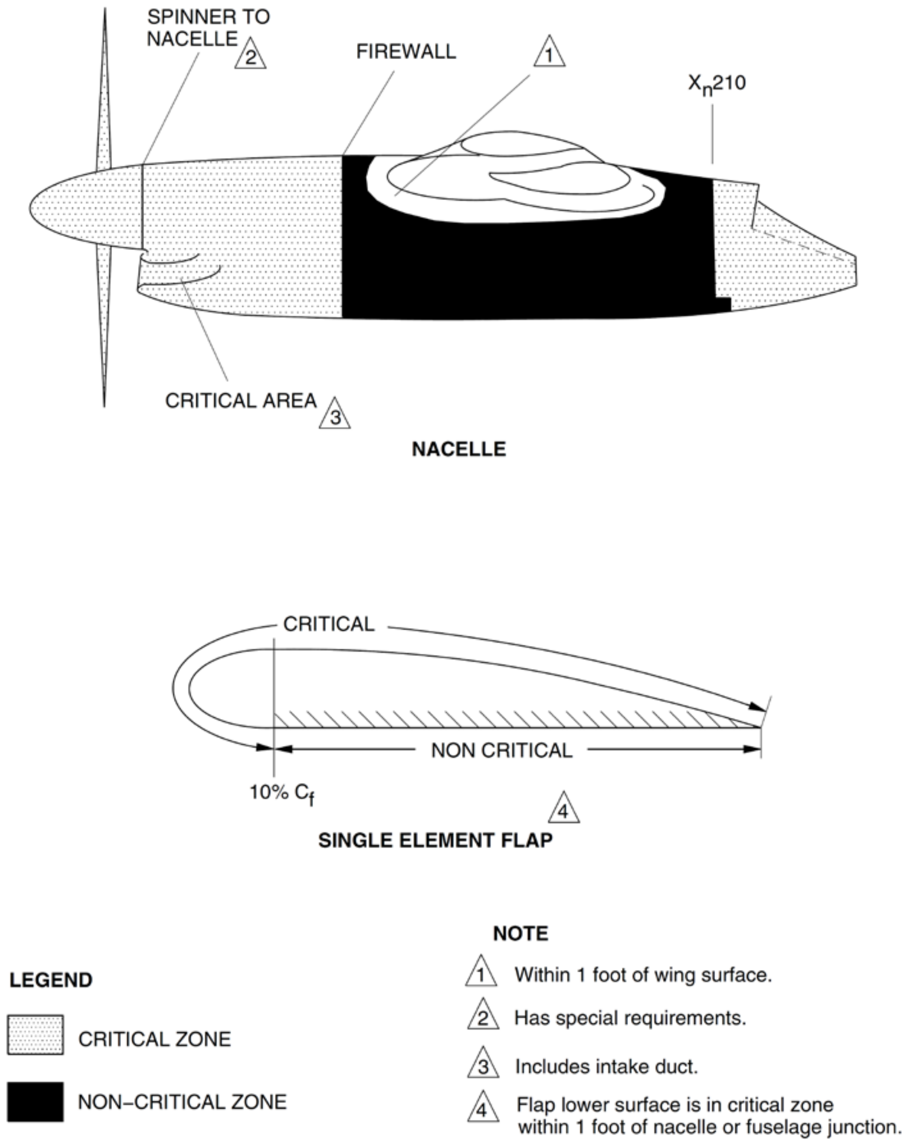


Figure 51-3 Critical and Non-Critical Zones - 1

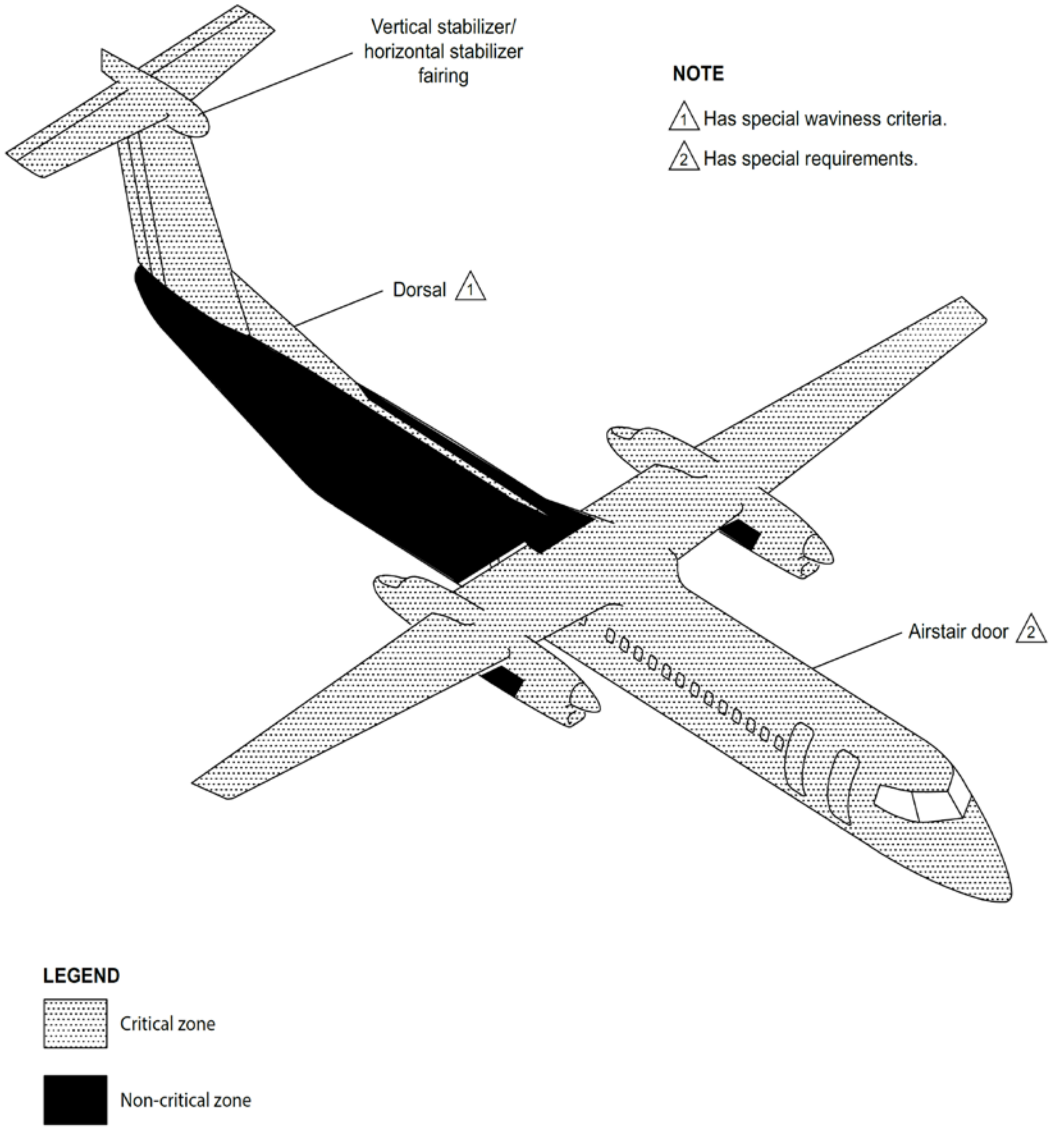


Figure 51-4 Critical and Non-Critical Zones-2

Bonding With Adhesive

General

Bonding or attaching different parts or materials to each other with adhesives.

Procedures

- Read and obey all instructions carefully
- Do not touch clean parts with bare hands. Use lint free cotton gloves
- Make sure that the shelf life of the adhesive materials have not expired
- Read and obey all manufacturer’s instructions.

NOTE: Porous surfaces may require two or more coats to ensure that sufficient adhesive remains on the surface. Allow each preceding coat to air dry before the next coat is applied.

NOTE: After adhesive has dried to a tack-free condition, parts which are not to be bonded immediately should be covered with clean Kraft paper.

Protective Paint Finishes

This section contains data that is necessary to apply and repair the aircraft protective paint finishes.

General Safety Precautions

- Obey all safety precautions and manufacturer’s instructions when you apply or repair primers and surface finishes.
- Use approved body protection suits, mask and breathing equipment.
- Spray room or shop must be equipped with exhaust system and be properly ventilated.
- No open flames, naked lights, infra−red or heat lamps in the paint area. Use approved explosion proof lighting and switches in the paint area.
- Should accidental eye contact occur with any chemicals or materials, flush with water and seek immediate medical attention.
- Special precautions must be taken when using materials containing isocyanates.
- Refer to the applicable manufacturer’s Material Safety Data Sheet (MSDS) for specific data on any of the materials used.

Environment Protection

- Many of the materials used can cause damage to the environment. When you dispose of these materials, you must follow the local government health and safety regulations.

Typical Sealing Methods

Refer Figures 51-5, 51-6, 51-7, 51-8 & 51-9

Typical sealing methods used with repairs to the aircraft structure are illustrated in figures 51-5 through 51-9.

Acceptable sealant use and locations to be used are listed in the SRM. Follow the instructions given by the manufacturer of the sealant as well as the instructions outlined in the AMM.

Types of Sealants:

- Fuel Tank Sealants
- Firewall Sealants
- Fuel Tank Corrosion Preventive Primer Sealant
- Aerodynamic and Pressure Sealing Sealants
- Faying Surface Sealant s for General Structure
- Alternate Fast Cure Sealants.

Refer to the Tables in the SRM for different sealants, manufacturers, specifications and applications.

NOTE: During sealing operations, make sure the drain holes in the structure remain unobstructed and free from sealant.

CAUTION: The sealant around the fuel tank dome anchor nuts are classified as Critical Design Configuration Control Limitation (CDCCL) items with the purpose of providing protection in the event of a lightning strike. The integrity of the sealant around the dome anchor nuts must be maintained to make sure that unsafe conditions do not develop by maintenance, modification or repairs. For details, refer to the SRM.

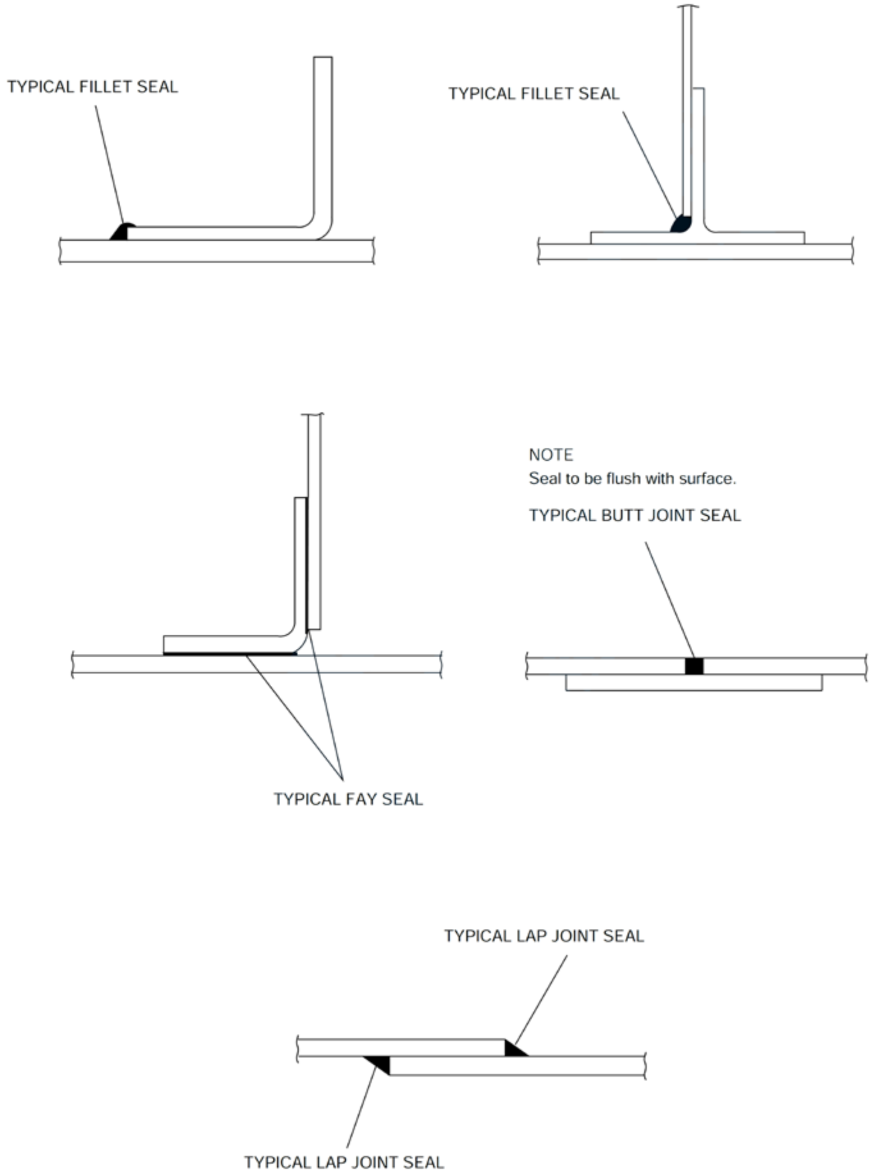


Figure 51-5 Typical Sealing Methods

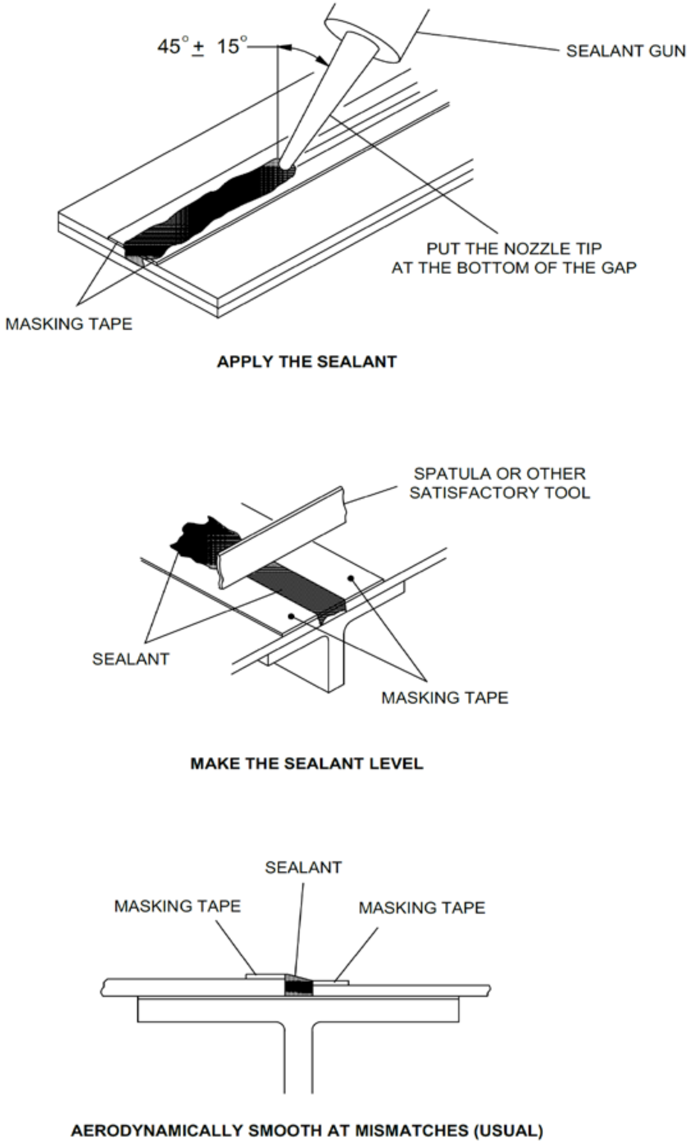


Figure 51-6 Sealing for Aerodynamic Smoothness

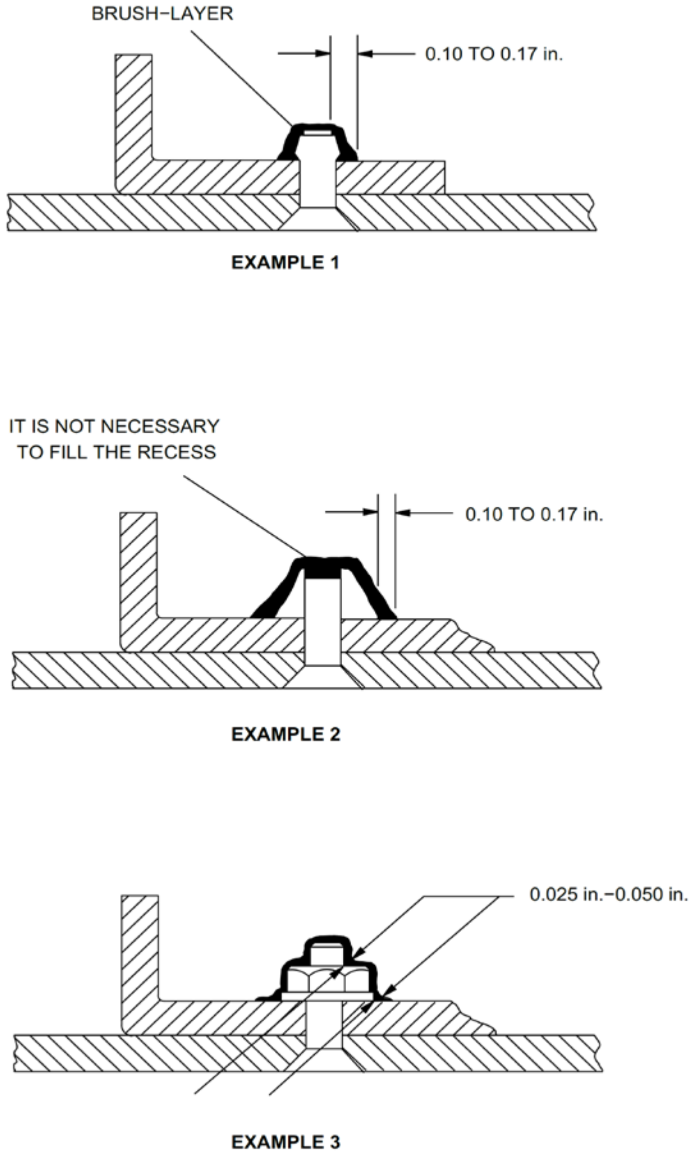


Figure 51-7 Sealant Application with a Brush

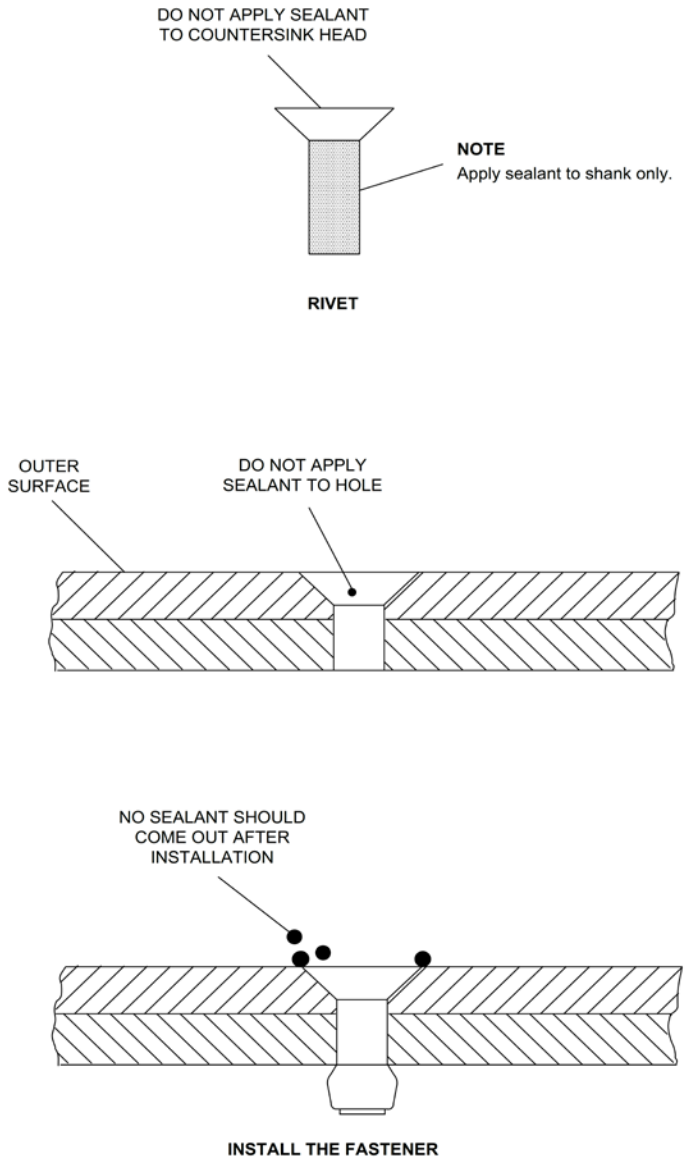
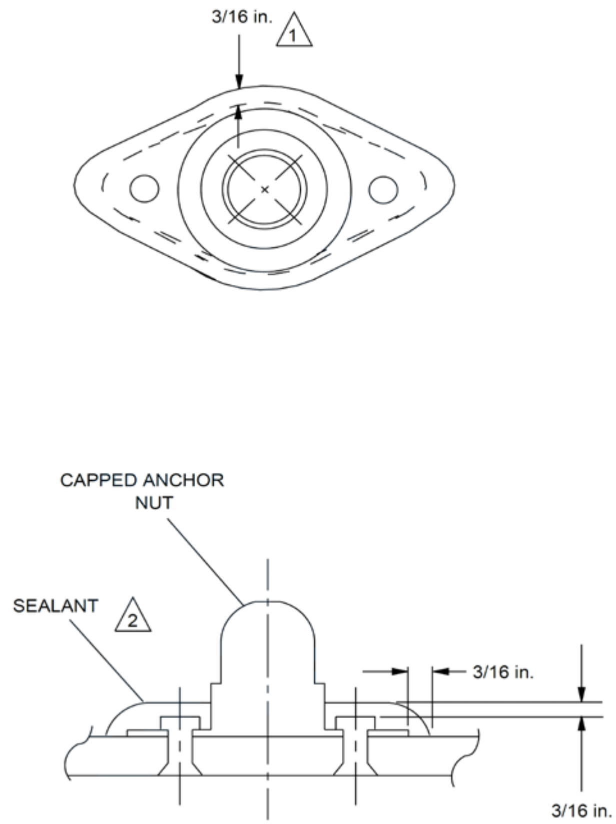


Figure 51-8 Wet Installation of Permanent Fasteners



NOTES

1. Sealant should extend 3/16 in. outward from the plate edge.
2. The sealant around the fuel tank access panel anchor nuts (Sta. Yw 42.000 to Yw 191.200) is classified as Critical Design Configuration Control Limitation (CDCCL) items with the purpose of providing protection in the event of the lightning strike. For the details refer to PSM 1−84−7, Part 2.

Figure 51-9 Dome Sealing Anchor Nut

Alternate Fast Cure Sealant

Alternate sealants can be used in the fuel cells when faster curing times are required to reduce the turn-around time on A.O.G aircraft. Time to fuel is based on sealant applied at 16 to 24°C. For every 12°C decrease in temperature, the time to fuel will double. Example: At 10°C a 6 hrs time to fuel is now 12 hrs. Above 29°C the application time of these materials is down to less than 15 minutes. The alternate sealant can be used where the original sealant is identified in the documentation.

NOTE: All De Havilland acceptable substitutes are based on Sealant Manufacturer's data or appropriate military specifications, See SRM for proper application.

CAUTION: The alternate sealants mentioned are not permitted for use in high temperature applications.

NOTE: When using some sealants (eg. Class B sealant) it is strongly recommended that the aircraft substrates be primed with "Adhesion Promoter".

CAUTION: Take care to make sure that sealant is applied within the time specified by the manufacturer of the sealant due to rapid cure time of some sealants. This is highly important when using "1/4" or "1/2" type of sealant.

NOTE: This figure is a general guideline on the curing behavior of sealants with respect to temperature variations. Wide variations in relative humidity (RH) will also affect curing times.

This guideline should only be applied in the range of 35% to 65% RH. Broken lines identify sealants that require an adhesion promoter when cure is forced over 90°F (32°C).

Dispatch of the Aircraft with Uncured or Missing Sealant

To allow aircraft to return to service prior to the completion of the curing cycle of Aerodynamic/Weather sealant applied to various surfaces.

Discussion

The application of High Speed Tape over the uncured sealant is acceptable, subject to the following:

Limitations

Tape may not be applied to the wing and horizontal stabilizer leading edges: If it loses adhesion during operation it could disturb the airflow, possibly changing the aircraft stall speeds.

Tape may not be used around engine air inlets, static ports, or NACA vents.

NOTE: It is permitted to operate the aircraft with the leading edge sealant removed from up to one wing leading edge, and one vertical stabilizer leading edge, and one horizontal stabilizer leading edge simultaneously from an aircraft.

NOTE: The above surfaces may be left unsealed for a maximum of 50 flight hours. At or prior to reaching the 50 hour limit, the leading edge(s) are to be removed; the areas must be inspected, cleaned, sealed and the sealant permitted to fully cure.

Recommendations

To improve the chances of the sealant fully curing:

- Seal with an approved sealant with the shortest cure time identified in the documents
- Allow the sealant to reach a “Tack Free” condition on the outer surface before tape application
- Ensure the area of adhesion for the tape is clean and dry
- Ensure the tape has properly adhered to the structure and component surfaces.
- Cover the exposed sealant with a suitable tape
- Keep the number of seams to a minimum and consider the airflow direction when applying the tape
- Tape is to be inspected daily for proper adhesion until removal
- If precipitation is encountered during flight, the adhesion should be confirmed prior to next flight
- Areas exhibiting signs of adhesion loss must be repaired prior to flight
- Tape is to be removed at first opportunity upon reaching final sealant cure
- The sealed area is to be checked to confirm sealant has remained in place and fully cured no later than the first ‘L’ check after the tape has been removed and flight operations resumed.

NOTE: Tape which comes in contact with ice crystals, snow, etc. may develop P-static build up resulting in noise on the radios and/or in the headsets.

NOTE: Due to temperature, pressure and humidity changes the sealant will encounter during aircraft operation, it is recommended the full cure time be extended by 50%.

Tapes

Aluminum Foil Tape: 3M 425 or 427, Permacell112, Mystic 7453 or Matrix Technology A662.

Aluminum Foil Tape with Conductive Adhesive: 3M 1170.

NOTE: P-static build up is less likely to develop using the Aluminum Foil conductive adhesive tape.

CAUTION: Aluminum Foil Tape is not suitable for around radiating devices such as V/Stab leading edges with HF antenna installation, radome or any antenna.

3M 8403, Matrix Technology M851, Airtech International Flashbreaker 1 or 2, or Richmond Aircraft Products Flashtape 1.

NOTE: Polyester/Polyvinyl Tape is suitable for all locations.

CAUTION: Polyester/Polyvinyl Tape is a possible source of P-static002E.

Solvent Cleaning

A. This topic specifies the requirements for solvent cleaning parts and surfaces. There are two types of cleaning, solvent washing or wiping.

NOTE: Protect all electrical equipments, plastics, rubber and other non−metallic materials before cleaning with solvents.

NOTE: Be careful when using solvents near windows or in the aircraft, most solvents and their vapors will cause crazing of acrylic surfaces.

B. When a particular solvent is not specified in any topic, refer to Table 1, for the type of solvent to be used for cleaning.

NOTE: Do not use chlorinated solvents such as Trichloroethane on Titanium or Titanium Alloy surfaces. Chlorinated solvents will cause hydrogen embrittlement on Titanium surfaces.

C. When cleaning thermoplastic material surfaces with solvents, do not allow the solvents to stay on the thermoplastic material surface longer than is necessary.

Safety Precautions

A. Keep solvents away from fire and other sources of ignition.

B. Provide sufficient ventilation when working in confined spaces, avoid prolonged breathing of solvent fumes. Wear an approved mask with air filters.

C. Do not smoke, eat or drink in areas where solvents are being handled. If ingestion of solvent occurs, get immediate medical attention.

D. Avoid skin contact with solvents. Wear protective gloves and safety glasses when handling or working with solvents.

E. Should accidental eye contact occur with solvents, wash thoroughly with water and get immediate medical attention.

F. Store cloths, soaked in solvent, in a sealed polyethylene bag when not in use.

G. Dispense solvent from polyethylene squeeze bottle or plunger−type containers which have been properly labelled with the solvent name.

H. Dispose of used wiper in oily waste cans.

I. Dispose of contaminated solvents and wiper cloths according to your local government health and safety regulations.

J. Refer to the applicable manufacturer’s Material Safety Data Sheet (MSDS) for specific safety data on any of the materials specified.

Cleaning Procedure

NOTE: Personnel responsible for cleaning manually with solvent or handling of solvent must have basic knowledge and familiarity with the procedures and requirements.

A. Degreasing

- Apply solvent (Reference Table 1) with a brush or cloth to the contaminated area.
- Rub the surface with a clean cloth or a soft bristle brush until the dirty material is removed.
- Flush the area with clean solvent.
- Dry the area with compressed air.

NOTE: Make sure the compressed air used for drying is free of water, oil or other contamination.

- Use clean, white, lint−free gloves to handle cleaned parts. Do not contaminate surface with bare hands.

B. Solvent Washing

- Wet the area to be solvent washed using a clean, lint−free wiping cloth soaked with the applicable solvent.
- Wipe the area dry with another clean, lint−free wiping cloth before the solvent evaporates.
- Use clean, white, lint−free gloves to handle cleaned parts. Do not contaminate surface with bare hands.
- Solvent washing with Dynamold DS 108F shall be done as follows:
  (a) Wet the area to be Dynamold DS 108F solvent washed using a clean, lint−free wiping cloth soaked with the solvent.

  (b) Allow the Dynamold DS 108F solvent to remain on the surface for 5 to 10 minutes before wiping dry with a clean, dry, lint−free wiping cloth.

  (c) Use clean, white, lint−free gloves to handle cleaned parts. Do not contaminate surface with bare hands.

C. Solvent Wiping

- Wipe the area to be solvent wiped using a clean, lint−free wiping cloth moistened with the applicable solvent.
- Wipe the area dry with another clean, lint−free wiping cloth before the solvent evaporates.
- Use clean, white, lint−free gloves to handle cleaned parts. Do not contaminate surface with bare hands.

NOTE: Solvent cleaned surface must be free of oil and grease. If oil and grease has not been removed completely after cleaning, clean again with solvent.

Storage

- Solvents shall be stored in their original containers and should be kept in flammable material storage cabinets (yellow).

Detergent Cleaning

A. This topic specifies the procedure for cleaning aircraft exterior surfaces with an alkaline water base cleaning compound.

NOTE: Do not clean the aircraft exterior surface when the temperature is less than 5°C (40°F). Ice can form.

B. Select the proper mixing ratios of compound to water for the type of cleaning required, normal, heavy soiled or area covered with exhaust product.

General Precautions

A. Use only cleaners with specifications, Mil–C–25769.

B. Do not let solvents stay on thermoplastic material surfaces for more time than it is necessary.

C. Make sure that materials such as paints, plastics and adhesives are fully cured before you clean them.

D. Make sure the compressed air used for solvent removal is free of water, oil and other contamination.

Safety Precautions

A. Use the correct maintenance stand when you clean the aircraft with alkaline cleaning solution.

B. Wear approved face mask when handling, mixing or working with alkaline solution.

C. Should eye contact with solution occur, wash thoroughly with water and seek immediate medical attention.

D. Avoid skin contact with alkaline solution. Wear protective gloves when handling, mixing or working with alkaline solution.

E. Refer to the manufacturer's Material Safety Data Sheet (MSDS) for specific information on the material specified.

F. Dispose of used or contaminated alkaline solutions according to your local government health and safety regulations.

Cleaning Procedure

A. Mixing Ratios.

(1) Normal Cleaning.

- Add one part of alkaline water base compound to ten parts of water.
- Mix thoroughly.

(2) Heavily Soiled Area.

- Add one part of alkaline water base compound to six parts of water.
- Mix thoroughly.

(3) Areas Covered with Exhaust Products.

- Add one part of alkaline water base compound to three parts of water.
- Mix thoroughly.

B. Preparation

- Inspect the surface to find the correct mixing proportions to be used.
- Make sure that all external compartments are closed and panels installed.
- All drain holes must be free from blockages.
- Install plugs and covers to prevent the entry of cleaning fluid into the internal areas of structures, systems and components. Eg: Engine intake and exhaust, Pitot/Static system etc.

C. Cleaning Procedure

- Apply water on all surfaces to be cleaned.

- With a spray equipment, mop or soft fibre brush, apply the alkaline cleaning solution to cover area to be cleaned.

NOTE: Do not apply the cleaning solution to very large areas where the solution may dry before flushing with water.

- Use a soft brush or mop and rub the surface with the wet solution.
- On heavy soiled areas, let the solution stay on the surface for 5 to 10 minutes and rub the surface with more force.

NOTE: Do not allow the solution to dry up. Continually keep the surface covered with cleaning solution. This prevents etched areas or unsatisfactory cleaning.

- Flush the cleaning solution from the surface with large quantities of clean water. Make sure no cleaning solution is left on the surface.

NOTE: Use hot water at 60°C (140°F) which will give the best results.

- Dry the surface with clean cloths or use an air gun.
- Remove all water that may be caught in crevices and openings.
- Repeat the cleaning procedure if surface is not clean.
- Remove all plugs and covers.

Cleaning of the Seat Rail

Maintenance Practices

Used to remove unwanted material so that the rails can be inspected.

CAUTION:

Painted and primed surfaces must be fully cured before the solvent cleaning.

Do not pour or spray solvent directly on parts with faying surfaces or entrapment areas.

Do not let the solvent enter surfaces or other entrapment areas. Use clean compressed air to remove the trapped solvents.

Epoxy and urethane paints and primers are considered to be “solvent resistant” and must be cleaned only with IPA or Naphtha.

Do not use MPK on “non solvent resistant” coatings.

Dispense Solvents from polyethylene squeeze bottles or plunger type containers, which have been properly labeled with the solvent name.

Tools and Equipment

- Vacuum
- Properly labeled polyethylene squeeze bottles
- Properly labeled plunger type containers.

Procedure

Use a vacuum and remove any loose unwanted material from the rail. Apply the de-greaser solvent to any unwanted material that remains on the rail as necessary and allow to soak. Use a Nylon brush or an abrasive pad and remove the unwanted material from the rail.

NOTE: Repeat steps until all unwanted material is removed from the rail.

Use a cloth and wipe dry the rail. Use a vacuum and remove any loose unwanted material from the rail. Remove tools, equipment, and unwanted materials from the work area.

Non-conductive finishes that you can remove:

Non-Conductive Finish	Base Metal	Color
Primers	All	Yellow or green
Topcoats	All	As specified
Anodic coating	Aluminium	Grey/yellow
Phosphate	Steel	Dull grey
Stains and dyes	All	Full range of colors
Fuel tank coating	All	Yellow or blue green
Adhesive on wire mesh	Aluminium wire	Not transparent

Non-conductive finishes that you cannot remove

Non-Conductive Finish	Base Metal	Color
Alodine 600, 1200	Aluminum	Iridescent gold or brown
Alodine 1000	Aluminum	Without color
Clear Iridite	Aluminum	Without color
Cadmium plate	Steel	Bright iridescent bronze
Chrome plate	Steel	Bright metallic
Nickel Plate	Steel	Satin chrome bright metallic
Tin Plate	Copper or Steel	Bright metallic

Temporary Allowance for Continued Operation with Damaged or Missing Erosion Protection Tape (PPT)

A. This topic provides a temporary allowance for continued operation with damaged or missing erosion protection tape (PPT) as applicable per Note mentioned below.

NOTE: This topic is applicable to aircraft with PPT installed per Modsum 4Q303300 and/or AMM TASK 51−25−61−390−801 (structure) and/or AMM TASK 51−25−61−390−803 (antennas), and is not applicable to the Polyurethane Protective Radome Boot (PPRB).

B. The temporary allowance involves removing any loose tape, ensuring the remaining tape is bonded to the structure and leaving the remaining tape as is for the temporary operating period.

NOTE: Erosion damage may occur during this temporary operating period to the unprotected composite surface where the tape is missing or has been removed.

C. There is no limit on the size of the missing tape, and this repair is valid for all PPT tape installed per Modsum 4Q303300 and/or AMM TASK 51−25−61−390−801 (structure) and/or AMM TASK 51−25−61−390−803 (antennas).

Temporary Allowance

A. This temporary allowance is valid for a maximum of 120 flight hours, at or before which time the tape is to be repaired or replaced.

Safety Precautions

A. Follow all the equipment manufacturer’s instructions.

B. Make sure that the aircraft is grounded.

C. Follow all the general shop safety precautions while doing the repair.

Procedure

A. Remove or trim off any loose tape and ensure all remaining tape is bonded to the structure. Use extreme care not to damage the underlying structure.

B. No loose edges or disbonded tape permitted for the temporary operating period.

C. No damage permitted to the underlying structure.

D. Leave the area as is for the temporary operating period

Resistance Measurement

General

A. The procedure that follows contains data necessary to do resistance measurements on aircraft structural components and aircraft components.

B. For the equipments that are installed on the aircraft, locations for the measurements are as follows:

1. For the Line Replaceable Units (LRUs) with an electrical connector, measure the resistance from the electrical connector to the electrically conductive surface on the aircraft structure.

2. For the LRUs without an electrical connector, measure the resistance from the unit to the electrically conductive surface on the aircraft structure.

3. For the hydraulic tubes, measure the resistance from the hydraulic tubes to the electrically conductive surface on the aircraft structure.

4. For the hydraulic equipment with no electrical connection, measure the resistance from the electrical conductive surface on the housing to the electrically conductive surface on the aircraft structure.

C. All the electrical equipment must be tested with the mating electrical connectors removed. The connectors must be installed after the test is completed.

Obey the safety precautions that follow:

Make sure that the aircraft is electrically grounded.

WARNING: OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

Procedure

A. Resistance Measurement of Conductive Coatings on Composite Structures

(1) You must do the resistance measurement of the component before you install it on the aircraft.

(2) Use a one mega ohm multimeter to measure the electrical resistance of the conductive layer as follows:

(a) Put the multimeter probes tightly against the conductive layer, approximately 12.00 in. (304.80 mm) apart.

NOTE: If the surface is less then 12.00 in. (304.80 mm) wide, then put the multimeter probes as far apart as possible.

(b) The maximum permitted resistance value for the conductive layer is 300,000 ohms.

NOTE: You must reject the component if the resistance measurement is larger than the maximum permitted value.

(c) Take a sufficient number of measurements to get a representative sample of the electrical resistance of the conductive layer.

B. Resistance Measurement Between the Conductive Layer on the Composite Structure and the Airframe

(1) Use a one megaohm multimeter to measure the electrical resistance between the conductive layer and the airframe as follows:

(a) Put one multimeter probe tightly against the conductive layer and the other probe tightly against the airframe.

(b) The maximum permitted resistance value is 300,000 ohms.

NOTE: You must reject the component if the resistance measurement is larger than the maximum permitted value.

(c) Take a sufficient number of measurements to get a representative sample of the electrical resistance of the conductive layer.

C. Resistance Measurement Between the De−Icer Boots and the Airframe

(1) Do the resistance measurement between the de−icer boots and the airframe as follows:

(a) Use a 500 V dc megger tester to measure the resistance between the de−icer boot and the airframe.

(b) Use a conductive tape approximately 0.5 in. by 2.0 in. (12.7 mm by 50.8 mm) to measure the electrical resistance of the de−icing boot.

NOTE: Solid copper electrode could be used as an alternative to the conductive tape.

(c) Put the conductive tape parallel to the edge of the de−icer boot, approximately 2.0 in. (50.8 mm) from the edge of the boot.

(d) The measured resistance must not be more than 100MΩ.

NOTE: You must reject the component if the resistance measurement is larger than the maximum permitted value.

Refer Figure 51-10

C. Resistance Measurement Between the Connectors and the Airframe

(1) Do the resistance measurement between the connectors and the airframe as follows:

(a) Use the bonding tester to measure the resistance between the connector and the primary structure.

(b) Put one of the bonding tester probe tightly against the connector and the other probe tightly against the primary structure. For the maximum permitted resistance value between the connectors and the airframe, refer to AMM Part II.

NOTE: You must reject the component if the resistance measurement is larger than the maximum permitted value.

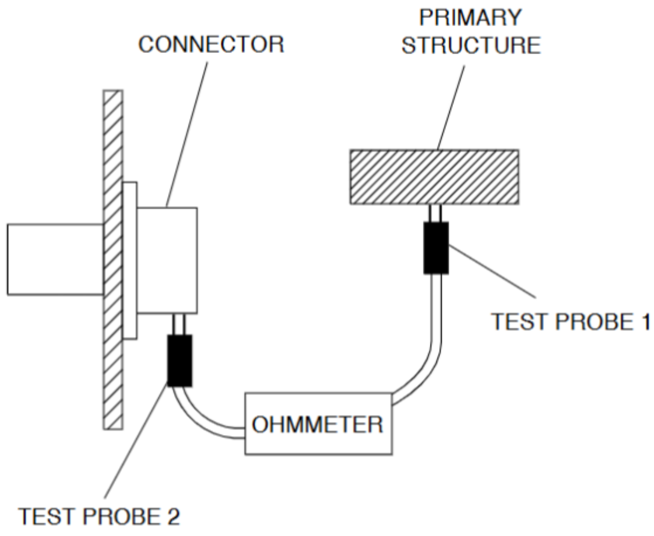


Figure 51-10 Measuring Resistance Between Connector and Structure

Refer Figure 51-11

D. Resistance Measurement Between Exposed Overbraid Shields and Connectors.

(1) Do the resistance measurement between exposed overbraid shields and connectors as follows:

(a) Use a multimeter to measure the resistance between the exposed overbraid shield and the connector.

(b) Put one multimeter probe tightly against the connector and the other probe tightly against the exposed overbraid shield. The maximum permitted resistance value between the connector and the exposed overbraid shield is 2.5 milliohms.

NOTE: You must reject the component if the resistance measurement is larger than the maximum permitted value.

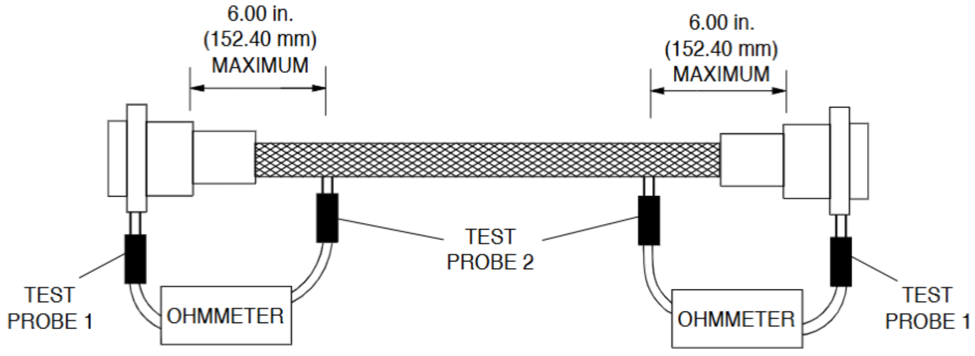


Figure 51-11 Measuring Resistance Between Overbraid Shield and Connector